LESSON 2

Balancing Redox Equations

THE OXIDATION-NUMBER METHOD AND THE HALF-REACTION METHOD

Standards

Reactions: apply systematic methods to balance oxidation-reduction equations.

Objectives

- How does the oxidation-number method balance a redox equation?
- How do you balance a net ionic redox equation in acidic or basic solution?
- How does the half-reaction method balance a redox equation?

Expectations

- Balance a redox equation using the oxidation-number method.
- Balance a net ionic redox equation in acidic or basic solution.
- Balance a redox equation by writing, balancing, and combining half-reactions.

Balancing Redox Equations

MAIN IDEA

Redox equations balance electrons lost against electrons gained, using either oxidation numbers or half-reactions.

KWS

WHAT I KNOW	WHAT I WANT TO KNOW	SOURCE
		Tyler DeWitt: How to Balance Redox Equations in Acidic Solution PhET: Balancing Chemical Equations

Vocabulary

Review

- oxidation
- reduction
- oxidation number
- oxidizing/reducing agent

New

- oxidation-number method
- net ionic equation
- half-reaction
- half-reaction method

The Oxidation-Number Method

- According to the oxidation-number method, chemical equations must be balanced to show the correct quantities of reactants and products.
- The number of electrons transferred from atoms must equal the number of electrons accepted by other atoms.
- The total increase in oxidation numbers must equal the total decrease in oxidation numbers in the reaction.

Table 4: The Oxidation-Number Method

1	**Assign oxidation numbers to all chemical elements in the equation.**
2	Identify the atoms or ions that are oxidized and reduced.
3	Determine the change in oxidation number for the atoms or ions that are oxidized and for the atoms or ions that are reduced.
4	Make the change in oxidation numbers equal in magnitude by adjusting coefficients in the equation.
5	If necessary, use the conventional method to balance the remainder of the equation.

Worked Example: Copper + Nitric Acid

- Balance the following redox equation using the oxidation-number method:

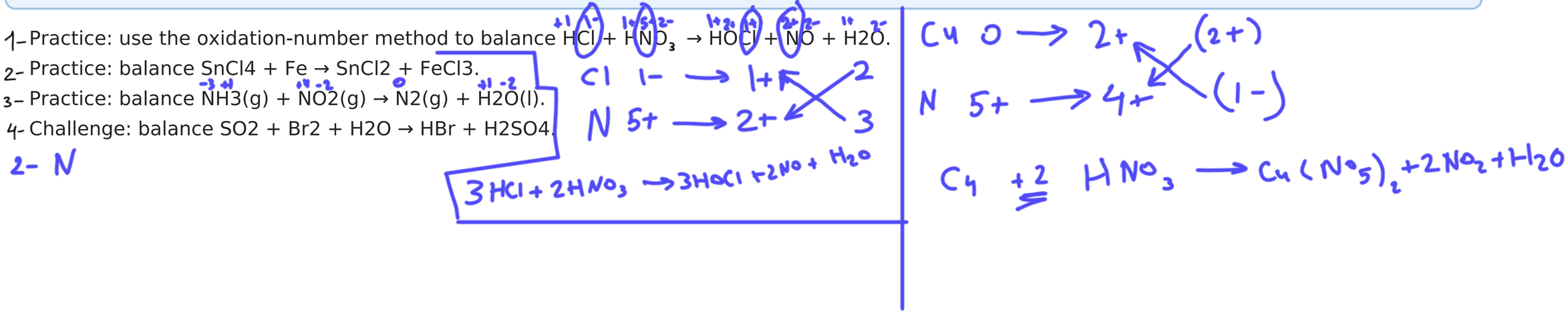
Cu + HNO3 → Cu(NO3)2 + NO2 + H2O

1- Practice: use the oxidation-number method to balance $HCl + HNO_3 \rightarrow HOCl + NO + H_2O$.

2- Practice: balance SnCl4 + Fe → SnCl2 + FeCl3.

3- Practice: balance NH3(g) + NO2(g) → N2(g) + H2O(l).

4- Challenge: balance SO2 + Br2 + H2O → HBr + H2SO4.



Balancing Net Ionic Redox Equations

- Sometimes it is preferred to express redox reactions in the simplest possible terms, showing only the oxidation and reduction processes.
- When balancing equations in acidic solution, hydrogen ions (H^+) or water molecules can be added to either side of the equation.
- When balancing equations in basic solution, hydroxide ions (OH^-) or water molecules can be added to either side of the equation.

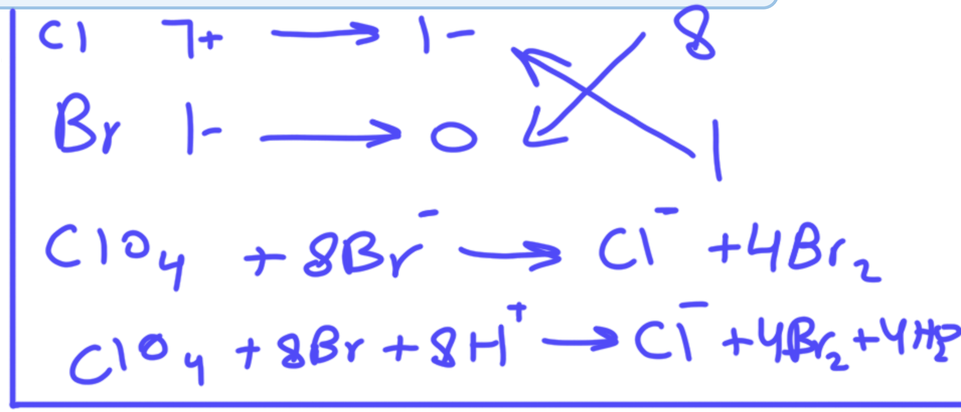
Molecular equation	**Cu(s) + 4HNO3(aq) → Cu(NO3)2(aq) + 2NO2(g) + 2H2O(l)**
Full ionic equation	$Cu(s) + 4H^+(aq) + 4NO3^-(aq) \rightarrow Cu^{2+}(aq) + 2NO3^-(aq) + 2NO2(g) + 2H2O(l)$
Net ionic (acidic solution)	$Cu(s) + 4H^+(aq) + 4NO3^-(aq) \rightarrow Cu^{2+}(aq) + 2NO3^-(aq) + 2NO2(g) + 2H2O(l)$

Balance a Net Ionic Redox Equation

- Balance the following redox equation:

$ClO4^-(aq) + Br^-(aq) \rightarrow Cl^-(aq) + Br2(g)$ (in acidic solution)

- Use the oxidation-number method to balance: $H2S(g) + NO3^-(aq) \rightarrow S(s) + NO(g)$ (in acidic solution).
- Balance: $Cr2O7^{2-}(aq) + I^-(aq) \rightarrow Cr^{3+}(aq) + I2(s)$ (in acidic solution).
- Balance: $Zn + NO3^- \rightarrow Zn^{2+} + NO2$ (in acidic solution).
- Challenge: balance $I^-(aq) + MnO4^-(aq) \rightarrow I2(s) + MnO2(s)$ (in basic solution).



Balancing Redox Equations Using Half-Reactions

- In chemistry, a species is any kind of chemical unit involved in a process.
- Oxidation-reduction reactions occur whenever a species that can give up electrons comes into contact with another species that can accept them.
- A half-reaction is one of the two parts of a redox reaction — the oxidation half or the reduction half.

Overall Reaction (unbalanced)	Oxidation Half-Reaction	Reduction Half-Reaction
$Fe + O2 \rightarrow Fe2O3$		$O2 + 4e^- \rightarrow 2O^{2-}$
$Fe + F2 \rightarrow FeF3$	$Fe \rightarrow Fe^{3+} + 3e^-$	$F2 + 2e^- \rightarrow 2F^-$
$Fe + HBr \rightarrow H2 + FeBr3$	(shared across rows)	$2H^+ + 2e^- \rightarrow H2$

Table 6: The Half-Reaction Method

1	**Write the unbalanced, net ionic equation for the reaction, omitting spectator ions.**
2	Write separate, incomplete equations for the oxidation and reduction half-reactions, including oxidation numbers.
3	Balance the atoms in the half-reactions; balance charges by adding electrons as reactants or products.
4	Adjust coefficients so the number of electrons lost in oxidation equals the number gained in reduction.
5	Add the half-reactions and cancel or reduce like terms on both sides.
6	Return spectator ions if desired; restore state descriptions.

Worked Example: Iron + Copper(II) Sulfate

Step	Result
1. Net ionic equation	$Fe + Cu^{2+} \rightarrow Cu + 2Fe^{3+}$
2. Half-reactions (unbalanced)	$Fe \rightarrow 2Fe^{3+}$ (oxidation, 0→+3) $Cu^{2+} \rightarrow Cu$ (reduction, +2→0)
3. Balance charge with electrons	$2Fe \rightarrow 2Fe^{3+} + 6e^-$ $Cu^{2+} + 2e^- \rightarrow Cu$
4. Equalize electrons (×3)	$2Fe \rightarrow 2Fe^{3+} + 6e^-$ $3Cu^{2+} + 6e^- \rightarrow 3Cu$
5. Add half-reactions	$2Fe + 3Cu^{2+} \rightarrow 3Cu + 2Fe^{3+}$
6. Restore spectator ions/states	2Fe(s) + 3CuSO4(aq) → 3Cu(s) + Fe2(SO4)3(aq)

Checkpoint: Balancing Redox Equations

- 1. Balance $KMnO_4(aq) + SnCl_2(aq) + HCl(aq) \rightarrow MnCl_2(aq) + SnCl_4(aq) + H_2O(l) + KCl(aq)$ using half-reactions.
- 2. Explain, in your own words, why the number of electrons lost must equal the number gained in a balanced redox equation.
- 3. Balance $Cr_2O_7^{2-}(aq) + I^-(aq) \rightarrow Cr^{3+}(aq) + I_2(s)$ (in acidic solution) using half-reactions. Leave the equation in ionic form.

Quick check — discuss with a partner, then confirm with your teacher.

Lesson Summary

Key Takeaways

- The oxidation-number method balances a redox equation by making the total increase and decrease in oxidation numbers equal.
- H^+ or H2O can be added in acidic solution; OH^- or H2O can be added in basic solution.
- The half-reaction method splits a redox equation into oxidation and reduction halves, balances each, then combines them.
- Electrons lost in oxidation must always equal electrons gained in reduction.

Vocabulary Recap

- oxidation-number method — balances using oxidation-number changes
- net ionic equation — shows only species that react
- half-reaction — the oxidation or reduction part of a redox reaction
- half-reaction method — balances each half, then combines them

Evaluation: Balancing Redox Equations

Complete independently. Review the summary if you get stuck, then check your answers with your teacher.

- 1. Use the oxidation-number method to balance: $NH_3(g) + NO_2(g) \rightarrow N_2(g) + H_2O(l)$.
- 2. Balance the net ionic equation $Zn + NO_3^- \rightarrow Zn^{2+} + NO_2$ (in acidic solution).
- 3. Use the half-reaction method to balance $Mn^{2+}(aq) + BiO_3^-(aq) \rightarrow MnO_4^-(aq) + Bi^{2+}(aq)$ (in acidic solution). Leave the equation in ionic form.
- 4. Explain the difference between the oxidation-number method and the half-reaction method — when might you prefer one over the other?
- 5. Challenge: Use the half-reaction method to balance $N_2O(g) + ClO^-(aq) \rightarrow NO_2^-(aq) + Cl^-(aq)$ (in basic solution).

Self-Evaluation Rubric

4 – Confident mastery • 3 – Solid grasp, minor gaps • 2 – Needs review • 1 – Revisit the lesson